

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
Community Water Infrastructure Reporting System
(CWIRS)
For Bamenda, Cameroon

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Prepared by: TEYIM YANICK NJIGMESHI

Prepared for: Academic Supervisor

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1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed overview of the **Community Water Infrastructure Reporting System (CWIRS)** for Bamenda, Cameroon. This document describes the project's features, stakeholders, functional and non-functional requirements, and system constraints.

CWIRS is a web-based platform that enables community members to report water service disruptions—specifically burst pipes and poor water quality—directly to water companies using live location mapping. The system aims to bridge the communication gap between residents and water service providers, leading to faster response times and improved water service management.

1.2 Document Conventions

Convention	Meaning
Must	Mandatory requirement
Should	Recommended but not mandatory
May	Optional feature
REQ-X	Functional requirement identifier
NFR-X	Non-functional requirement identifier

1.3 Intended Audience and Reading Suggestions

Audience	Purpose	Recommended Sections
Academic Supervisor	Project approval and evaluation	Full document

Developer (You)	Implementation guidance	Sections 3, 4, 6
Water Company Stakeholders	Understanding system capabilities	Sections 2, 3, 7

1.4 Product Scope

CWIRS is a web-based application designed specifically for Bamenda, Cameroon. The system:

- **Accepts reports** from community members regarding burst pipes, poor water quality, no water flow, and other water service disruptions
- **Captures live location** (GPS coordinates) automatically from the reporter's device
- **Displays all reports** on an interactive map for easy visualization
- **Provides water companies** with a dashboard to view, acknowledge, and resolve reported issues
- **Enables report tracking** using unique identification numbers
- **Stores historical data** for future analysis and accountability reporting

The system is **not** a real-time water quality testing platform. It relies on community-submitted observations rather than sensor data.

1.5 References

Reference	Source
IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications	IEEE
WHO Guidelines for Drinking-water Quality (2024)	World Health Organization
Cameroon Water Utility Regulatory Standards	Ministry of Water Resources, Cameroon

2. Overall Description

2.1 Product Perspective

CWIRS is a **new, standalone system** being developed from scratch. It does not replace any existing system in Bamenda because no centralized water disruption reporting platform currently exists.

2.2 Product Functions

REQ Overview: The system will perform the following high-level functions:

Function ID	Function Description
F-01	Accept and store water disruption reports from community members
F-02	Automatically capture and store reporter's GPS location
F-03	Display all active and resolved reports on an interactive map
F-04	Provide water company staff with a dashboard to manage reports
F-05	Allow water companies to update report status (Pending → Acknowledged → Resolved)
F-06	Enable photo upload as evidence for reports
F-07	Provide unique tracking IDs for each report
F-08	Allow community members to check report status using tracking ID
F-09	Permit upvoting of existing reports to indicate severity/popularity
F-10	Export report data to CSV for analysis

2.3 User Characteristics

User Type	Technical Skill	Frequency of Use	Primary Goal
Community Member (Reporter)	Low (basic phone/computer use)	Occasional (as issues arise)	Report water problems quickly
Community Member (Viewer)	Low	Frequent (checking status)	Track issue resolution
Water Company Staff	Medium	Daily	Manage and respond to reports
System Administrator	High	Weekly	User management, system oversight

2.4 Operating Environment

Component	Environment
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Client (User)	Web browser: Chrome (latest), Firefox (latest), Safari (latest); Mobile browsers supported
Server	Linux (Ubuntu 20.04+) or Windows Server; Apache/Nginx
Database	MySQL 5.7+ or PostgreSQL 12+
Hosting	Local server (XAMPP/WAMP for development) or cloud hosting (Render, PythonAnywhere, or similar)

2.5 Design and Implementation Constraints

Constraint	Description
C-01	System must be accessible on low-bandwidth connections (2G/3G networks common in Bamenda)
C-02	System must function on low-end Android phones and basic feature phones (via mobile web)
C-03	No payment gateway required (academic project)
C-04	No real-time SMS integration required (out of scope due to cost)
C-05	Maps must work offline or with cached tiles where possible

2.6 Assumptions and Dependencies

Assumptions:

ID	Assumption
A-01	Users have access to a smartphone or computer with internet connection
A-02	Users' devices have GPS capability (for smartphones)
A-03	Water company staff have basic computer literacy
A-04	Water companies will check the dashboard at least twice daily

Dependencies:

ID	Dependency
D-01	System depends on free map service (Leaflet.js with OpenStreetMap)
D-02	System depends on web hosting service for deployment
D-03	System depends on GPS availability on user devices

3. System Features and Requirements

3.1 User Reporting Feature

Description: Community members can submit a report describing a water service disruption.

REQ-01: Report Submission Form

Requirement ID	Requirement Description	Priority
REQ-01.1	System must provide a web form for submitting water disruption reports	High
REQ-01.2	Form must include fields: Issue Type, Description, Optional Photo	High
REQ-01.3	Issue Type must be a dropdown with options: "Burst Pipe," "Poor Water Quality," "No Water Flow," "Other"	High
REQ-01.4	Description field must accept up to 500 characters	Medium
REQ-01.5	Reporter name field must be optional (anonymous reporting allowed)	Medium

REQ-02: Report Submission Processing

Requirement ID	Requirement Description	Priority
REQ-02.1	System must generate a unique tracking ID for each report (format: WB-YYYYMMDD-XXXX)	High
REQ-02.2	System must record timestamp of submission automatically	High
REQ-02.3	System must display the tracking ID to the user after successful submission	High
REQ-02.4	System must store all report data in database	High

3.2 Live Location Mapping Feature

Description: The system automatically captures the reporter's geographic location.

REQ-03: Location Capture

Requirement ID	Requirement Description	Priority
REQ-03.1	System must request user's location permission via browser geolocation API	High

REQ-03.2	System must automatically capture latitude and longitude coordinates upon permission grant	High
REQ-03.3	System must display a draggable map pin allowing users to adjust location if GPS is inaccurate	Medium
REQ-03.4	System must store latitude and longitude as decimal values (precision: 6 decimal places)	High
REQ-03.5	System must handle cases where location permission is denied (allow manual address entry)	Medium

3.3 Interactive Map Visualization Feature

Description: All reported issues are displayed on an interactive map.

REQ-04: Map Display

Requirement ID	Requirement Description	Priority
REQ-04.1	System must display an interactive map (using Leaflet.js or similar) on the main page	High
REQ-04.2	Map must show markers for all reported issues	High
REQ-04.3	Map markers must be color-coded by status (Red = Pending, Yellow = Acknowledged, Green = Resolved)	High
REQ-04.4	Clicking a marker must display a popup with issue summary (type, status, time reported)	High
REQ-04.5	Map must center on Bamenda, Cameroon (coordinates: 5.9339° N, 10.1568° E) by default	High
REQ-04.6	Map should allow zooming from street level to city level	Medium

3.4 Water Company Dashboard Feature

Description: Water company staff can view and manage reported issues.

REQ-05: Dashboard Access

Requirement ID	Requirement Description	Priority
REQ-05.1	System must provide a secure login page for water company staff	High
REQ-05.2	System must restrict dashboard access to authenticated users only	High

REQ-05.3	Dashboard must display a table of all reports sorted by date (newest first)	High
REQ-05.4	Dashboard must display summary statistics (total reports, pending, resolved)	Medium
REQ-05.5	Dashboard must allow filtering by status, date range, and issue type	Medium

REQ-06: Report Management

Requirement ID	Requirement Description	Priority
REQ-06.1	System must allow water company staff to change report status (Pending → Acknowledged → Resolved)	High
REQ-06.2	System must record the timestamp of each status change	Medium
REQ-06.3	System must allow water company staff to add internal notes to a report	Medium
REQ-06.4	System must allow water company staff to assign a repair team name to a report	Low

3.5 Report Tracking and Status Update Feature

Description: Community members can check the status of their reports.

REQ-07: Public Status Check

Requirement ID	Requirement Description	Priority
REQ-07.1	System must provide a "Check Report Status" page accessible without login	High
REQ-07.2	Page must have a search box for entering tracking ID	High
REQ-07.3	System must display report details (issue type, location, status, timestamp, resolution notes if any)	High
REQ-07.4	System must show an error message if tracking ID is not found	Medium

3.6 Photo Upload Feature

Description: Users can attach photos as evidence.

REQ-08: Photo Handling

Requirement ID	Requirement Description	Priority
REQ-08.1	System must allow users to upload up to 3 photos per report	Medium
REQ-08.2	System must accept image formats: JPG, PNG, JPEG	Medium
REQ-08.3	System must limit file size to 5 MB per photo	Medium
REQ-08.4	System must store photos securely and display them on report details page	Medium
REQ-08.5	System must compress images to reduce storage and bandwidth usage	Low

3.7 Upvote/Severity Voting Feature

Description: Community members can upvote existing reports to indicate urgency.

REQ-09: Voting System

Requirement ID	Requirement Description	Priority
REQ-09.1	System must display an upvote button on each report (public view)	Medium
REQ-09.2	System must prevent a single user from upvoting the same report multiple times	Medium
REQ-09.3	System must sort reports by upvote count on the water company dashboard (highest first)	Medium
REQ-09.4	System must display upvote count on map popups	Low

3.8 Historical Data Export Feature

Description: Administrators can export report data for analysis.

REQ-10: Data Export

Requirement ID	Requirement Description	Priority
REQ-10.1	System must allow administrators to export all report data to CSV format	Medium
REQ-10.2	Export must include: tracking ID, issue type, coordinates, status, timestamp, resolution time	Medium
REQ-10.3	System must allow filtering by date range before export	Low

REQ-10.4	Exported data must be usable for future analysis (including machine learning if desired)	Medium
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4. External Interface Requirements

4.1 User Interfaces

Interface	Description
Public Homepage	Map view with markers; buttons for "Report Issue" and "Check Status"
Report Form Page	Form with fields, location capture, photo upload, submit button
Status Check Page	Input field for tracking ID; displays report details
Water Company Login Page	Username/password form
Water Company Dashboard	Table of reports, summary stats, status update buttons
Admin Panel	User management, system settings, data export

4.2 Hardware Interfaces

Interface	Requirement
GPS Receiver	Browser geolocation API (uses device GPS if available)
Camera	Browser file input for photo upload (uses device camera on mobile)
Display	Minimum resolution 800x600; responsive design for mobile (320px width)

4.3 Software Interfaces

External Software	Purpose	Interface Method
OpenStreetMap (via Leaflet.js)	Display interactive maps	REST API / JavaScript library
Browser Geolocation API	Capture user location	JavaScript API
MySQL/PostgreSQL	Data storage	Database driver (PDO, mysqli)
Web Server (Apache/Nginx)	Serve application	HTTP/HTTPS

4.4 Communication Interfaces

Protocol	Use
HTTP/HTTPS	All client-server communication
JSON	Data exchange format (if API implemented)
TCP/IP	Underlying network communication

5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement	Target
NFR-01	Page load time (homepage)	< 3 seconds on 3G connection
NFR-02	Report submission processing time	< 5 seconds
NFR-03	Map marker rendering (up to 500 markers)	< 2 seconds
NFR-04	Concurrent users supported	Minimum 50

5.2 Security Requirements

ID	Requirement
NFR-05	All passwords must be hashed (bcrypt or similar) before storage
NFR-06	Dashboard access must require authentication
NFR-07	All data transmission must use HTTPS in production
NFR-08	Photo uploads must be scanned for malicious content (basic validation)
NFR-09	SQL injection prevention: use parameterized queries or ORM

5.3 Usability Requirements

ID	Requirement
NFR-10	System must be usable on mobile phones with screens as small as 320px width
NFR-11	All forms must have clear labels and validation error messages
NFR-12	Map interface must be intuitive (zoom, pan, click markers)

NFR-13	Report submission must take less than 2 minutes for a typical user
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5.4 Reliability Requirements

ID	Requirement
NFR-14	System must handle invalid user inputs gracefully (no crashes)
NFR-15	System must not lose data if a submission fails (validation before processing)
NFR-16	Database backups must be possible (manual export)

5.5 Availability Requirements

ID	Requirement
NFR-17	System should be available 24/7 during demonstration period
NFR-18	Planned maintenance must be announced (for real deployment)

5.6 Scalability Requirements

ID	Requirement
NFR-19	Database schema must support future expansion (new report types, user accounts)
NFR-20	System architecture should allow future addition of SMS reporting

6. Data Requirements

6.1 Data Model

Core Entities:

- **Report** – Core entity representing a water disruption report
- **User** – Registered users (water company staff, administrators)
- **Photo** – Images attached to reports
- **Status History** – Log of status changes for each report

6.2 Database Schema

Table: reports

Column Name	Data Type	Constraints	Description
Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique internal ID

tracking_id	VARCHAR(20)	UNIQUE, NOT NULL	User-facing tracking ID
issue_type	ENUM	NOT NULL	Burst pipe, Poor quality, No flow, Other
Description	TEXT		User description (max 500 chars)
Latitude	DECIMAL(10,8)	NOT NULL	GPS latitude
Longitude	DECIMAL(11,8)	NOT NULL	GPS longitude
Address	VARCHAR(255)		Human-readable address (optional)
reporter_name	VARCHAR(100)		Optional, nullable
upvote_count	INT	DEFAULT 0	Number of upvotes
Status	ENUM	DEFAULT 'pending'	pending, acknowledged, resolved
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Submission time
acknowledged_at	TIMESTAMP	NULL	When status changed to acknowledged
resolved_at	TIMESTAMP	NULL	When status changed to resolved
assigned_team	VARCHAR(100)		Repair team name (optional)
internal_notes	TEXT		Water company private notes

Table: users

Column Name	Data Type	Constraints	Description
Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user ID
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	bcrypt hash

role	ENUM	NOT NULL	'water_company', 'admin'
company_name	VARCHAR(100)		For water company users
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

Table: photos

Column Name	Data Type	Constraints	Description
Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique photo ID
report_id	INT	FOREIGN KEY (reports.id)	Associated report
file_path	VARCHAR(255)	NOT NULL	Storage path
uploaded_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload time

Table: upvotes

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID
report_id	INT	FOREIGN KEY (reports.id)	Upvoted report
session_id	VARCHAR(255)	NOT NULL	To prevent duplicate upvotes
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upvote time

6.3 Data Validation Rules

Field	Validation Rule
Latitude	Must be between -90 and 90
Longitude	Must be between -180 and 180
Issue type	Must be one of the defined options

Description	Max 500 characters; no HTML/script tags
Photo	Max 5 MB; allowed types: JPG, PNG, JPEG
Tracking ID	Format: WB-YYYYMMDD-XXXX (auto-generated)

7. User Roles and Permissions

Role	Permissions
Guest (Public)	View map, submit reports, check report status, upvote reports
Water Company Staff	Login, view dashboard, see all reports, update status, add internal notes
Administrator	All water company permissions + manage users, export all data, system configuration

Permission Matrix:

Action	Guest	Water Company	Admin
View map	✓	✓	✓
Submit report	✓	✓	✓
Check status	✓	✓	✓
Upvote report	✓	✓	✓
View dashboard	✗	✓	✓
Update report status	✗	✓	✓
Add internal notes	✗	✓	✓
Manage users	✗	✗	✓
Export data	✗	✗	✓

8. Appendices

Appendix A: Glossary

Term	Definition
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Burst Pipe	A broken water pipe causing leakage or flooding
CWIRS	Community Water Infrastructure Reporting System
GPS	Global Positioning System – for capturing location coordinates
Issue Type	Category of water disruption (burst pipe, poor quality, no flow, other)
Leaflet.js	Open-source JavaScript library for interactive maps
OpenStreetMap	Free, editable map of the world used as map source
Pending	Initial status – report submitted, not yet acknowledged
Acknowledged	Water company has seen the report and is investigating
Resolved	Issue has been fixed by water company
Tracking ID	Unique identifier for each report (e.g., WB-20260515-0001)
Upvote	User action to indicate agreement or urgency of a report

End of Software Requirements Specification Document